

TrES-2b

R-2040

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_121020 · created 2026-08-08T22:51:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456220.71717 +/- 0.00018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.061 +/- 0.04
Transit depth (Rp/Rs)^2	0.38 +/- 0.49 [%]
Orbital Inclination (inc)	88.5 +/- 3.17
Airmass coefficient 1 (a1)	1.1026 +/- 0.0066
Airmass coefficient 2 (a2)	-0.0684 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	0.59 %
Best Comparison Star	#3 - [292, 205]
Optimal Aperture	3.59
Optimal Annulus	3.59
Transit Duration (day)	0.056 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.061 ± 0.04 vs 0.1254 (deviation 1.43σ)
Duration fit vs published	0.056 d vs 0.0746 d (deviation 1.5σ)
Mid-time O–C	-0.3 min
Depth SNR	1.4
Residual scatter	0.59 %

Light curve

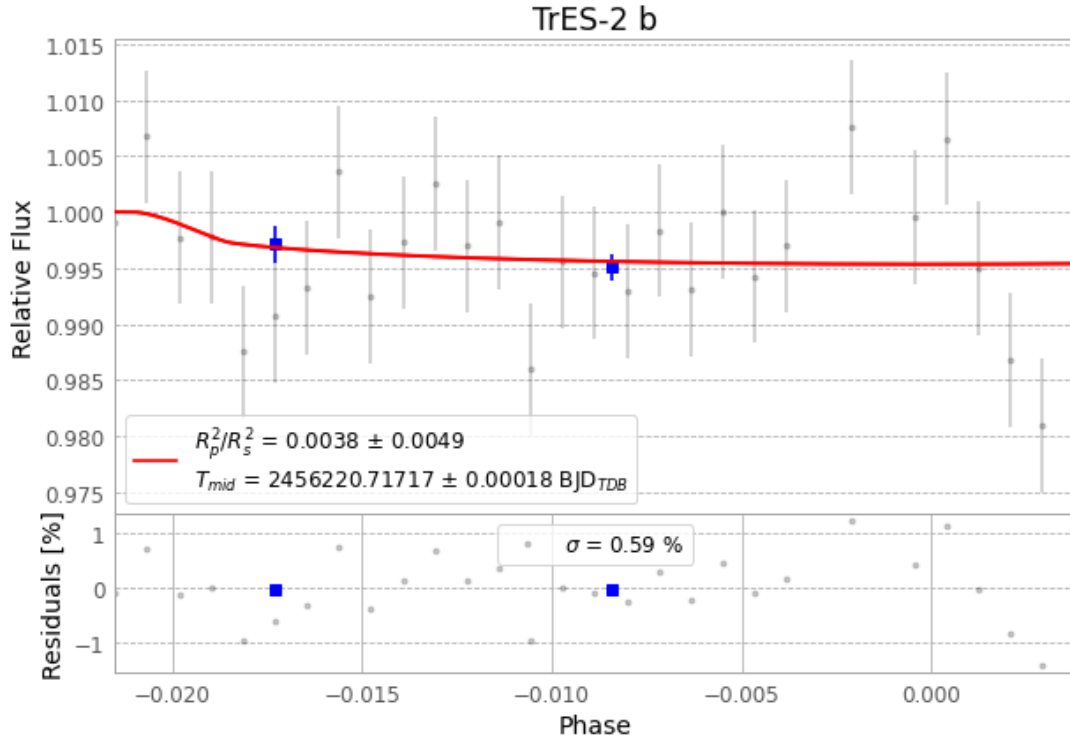


Figure 1 — FinalLightCurve_TrES-2 b_2012-10-19.png (SHA-256 675388fafdb801ae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 912eba27ecfa9c10...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

31 FITS frames (20 MB), TRES-2121020035412.FITS → TRES-2121020052412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-121020.log (81aa4e2720e9...), FinalLightCurve_TrES-2 b_2012-10-19.png (675388fafdb8...), FinalLightCurve_TrES-2 b_2012-10-19.csv (468aafb4c0ae...)

Compute resources

Wall time 125.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 22:51Z.